

# Participant Behavioral Patterns in a Citizen Health-Promotion Program

Large-scale step-count evidence from Seoul's *Wrist Doctor 9988*

---

Youngdoo Jeong<sup>1</sup>   Kione Kim<sup>2</sup>   Yonghee Lee<sup>1</sup>

JSM 2026 • Boston • August 4, 2026

<sup>1</sup>Department of Statistics, University of Seoul   <sup>2</sup>Urban Big Data and AI Institute, University of Seoul

## 1. The program

- Why population-scale activity programs
- Seoul's *Wrist Doctor 9988*
- Enrollment phases & how points are earned

## 2. Data & metrics

- 2.18M members
- Engagement metrics  
(participation, goal, active, retention — our definitions)

## 3. Results

- Who participates
- Steps by gender, age, day of week
- Engagement & retention over time
- Regional variation

## 4. Findings & implications

# The program

---

## Background: physical inactivity at population scale

- Physical inactivity is a leading, persistent risk factor for non-communicable disease worldwide.
- **Mobile-health (mHealth)** programs — wearables, an app, step goals, and financial rewards — aim to raise activity at the population level.
- Real-world evidence *at scale* is still scarce; most studies are small or short.
- **Our setting:** a large, continuously running *municipal* program, observed for nearly four years.

### This talk

We characterise **who joins**, **how they move**, and **who keeps going** in a 2.18M-member program — defining engagement metrics for which prior literature offers little precedent.

Analytic framing follows large mHealth evaluations such as Singapore's National Steps Challenge (Yao et al., 2022).

## Seoul's *Wrist Doctor 9988*: a city-wide health program

- Smart-healthcare initiative of the **Seoul Metropolitan Government**; launched 2021.
- Name = *“live to 99, stay active to 88”* — the goal is durable healthy habits, not a one-off challenge.
- Each participant gets a **smart band** (or links their own watch / smartphone) and a dedicated app that records daily steps.
- Activity is converted into **points**, redeemable as local-currency rewards.
- Run continuously since the pilot, with periodic enrollment waves.

### Pilot evaluation (Phase 1, 50k)

- 97% completed the program
- 64% recorded data weekly+
- 85% reported better habits
- 45% increased their steps

Source: Seoul Metropolitan Government.

## Five enrollment phases (and counting)

| Phase | Started    | New recruits | Ages  | Note                                               |
|-------|------------|--------------|-------|----------------------------------------------------|
| P1    | Nov 2021   | 50,000       | 19–64 | Pilot (“On-Seoul Health-On”); smart bands provided |
| P2    | Dec 2022   | 180,000      | 19–69 | Renamed <i>Wrist Doctor 9988</i> (Jun 2022)        |
| P3-1  | Aug 2023   | 150,000      | 19–75 | + mental-care, home-training                       |
| P3-2  | Nov 2023   | 70,000       | 19–75 | Additional recruitment wave                        |
| P4    | Mar 2024 → | rolling      | 19+   | Smartphone-only; year-round; → 2.18M cumulative    |

**What this means for the data:** the panel is *not* a single cohort — successive waves enter with different eligibility and devices, and Phase 4 enrolls continuously.

Source: Seoul Metropolitan Govt. policy archive. Recruit counts are announced targets; members with usable step data differ.

# How points are earned

## Core: the daily step goal

- **8,000 steps/day** → **200 points**  
(ages 70+: 5,000 steps; or 200 kcal via the app's exercise mode)
- This daily goal is the primary unit of activity in the program.

## Weekly & streak bonuses

- Goal on  **$\geq 3$  days in a week** → a **weekly bonus** (e.g. +500 pts for activity, on top of the daily points).
- Sustaining that for **4 consecutive weeks** → a further bonus (e.g. +800 pts).

## Other point-earning activities

- Diet / meal logging in the app
- Health quizzes (e.g. 100 pts) & card-news education
- Survey & health-info consent at sign-up
- Designated-route / mission walks

## Key analytic choice

The program itself rewards the  **$\geq 3$  days/week** threshold — so we adopt it to define an “**active**” participant.

## Redeeming points

- Up to **100,000 points** per person;  
**1 point = 1 KRW**.
- Converted to **Seoul Pay** (local mobile money).
- Spendable at convenience stores, pharmacies, clinics, and other affiliated merchants.
- Redemption windows are announced each cycle.

## Devices & data capture

- HPB-style **smart band** provided in early phases; from P4, **smartphone-only** participation.
- **iPhone** ↔ Apple Health;  
**Galaxy** ↔ Samsung Health.
- Steps sync to the app and upload to the program server.
- Eligibility: Seoul residents (incl. workers / students), age threshold by phase.

Source: Seoul Metropolitan Government program documentation. Point caps / windows reflect the studied period and may change.

## Data & metrics

---

- Program-derived **summary statistics** (de-identified), **Nov 2021 – Jul 2025**.
- Per-member attributes: **sex**, **age band**, **district (gu)**, **enrollment type**, plus daily step counts.
- Two populations we refer to throughout:
  - **2.18M** cumulative registrations
  - **1.87M** analysis cohort  
(members with sufficient step history to be classified)
- Tens of millions of person-days of step records.

2.18 M

cumulative members

1.87 M

analysis cohort

### Scope of this study

We characterise **uptake, engagement, and behavioral patterns**. Causal **effectiveness** (pre/post step change) is left to future work.

## Engagement metrics we define

- **Participation rate.** Share of eligible members who *sync at least once* in the week — i.e. the app records any activity for them. The broadest signal: “did they show up?”
- **Goal-achievement rate ( $\geq 1\times/\text{wk}$ ).** Share who *meet the daily step goal* (8,000 steps; 5,000 for ages 70+) on *at least one* day. Stricter than participation — they actually hit the target, not just opened the app.
- **Active rate ( $\geq 3\times/\text{wk}$ ).** Share who meet the goal on *three or more* days in the week. Our headline measure of *sustained* weekly activity.
- **Retention at week  $k$ .** Of members who joined  $k$  weeks ago, the share *still recording* — a survival view of a joining cohort, not a single-week snapshot.

### Why this matters

These weekly rates are **our own constructions**; prior mHealth work rarely formalises a *sustained*-activity measure.

- The three weekly rates are **nested**: every active ( $\geq 3\times$ ) member also achieved the goal ( $\geq 1\times$ ) and participated — so  $\text{Active} \leq \text{Goal} \leq \text{Participation}$  by construction.
- The  **$\geq 3$  days** cut is **not arbitrary**: it matches the program’s own **weekly bonus** rule, making “active” a policy-meaningful threshold.

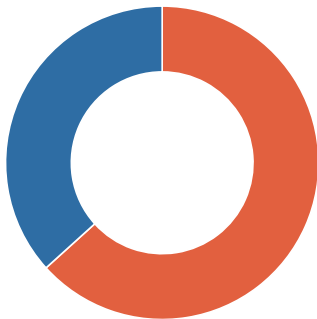
## Results

---

## Who participates: women sign up more, ages skew middle-aged

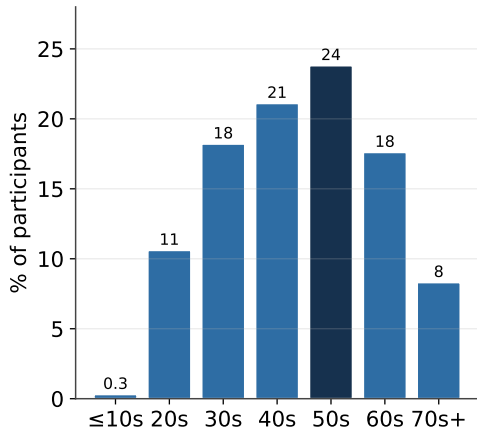
### Participant profile (analysis cohort, N = 1.87M)

#### Gender



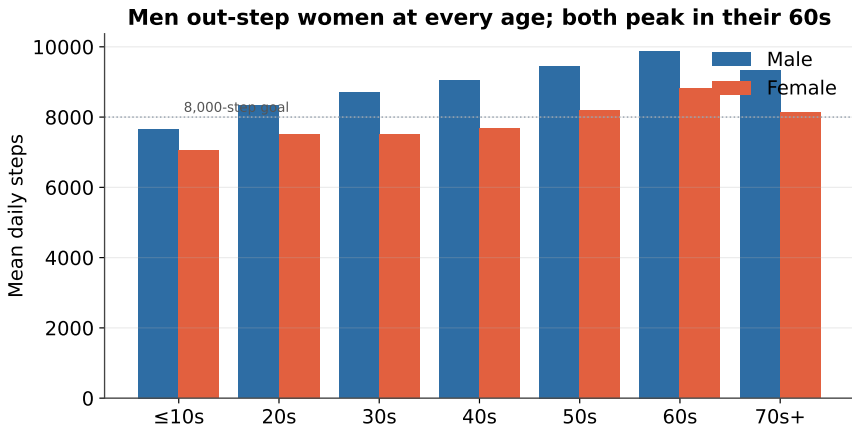
Female 63.3%  
Male 36.7%

#### Age group



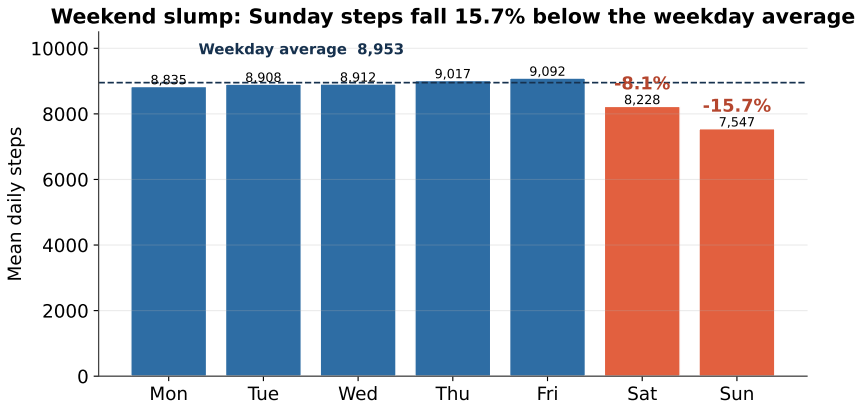
**63%** of members are women; participation peaks in the **40s–50s**, with strong representation into the 60s.

## Daily activity: men out-step women; both peak in their 60s



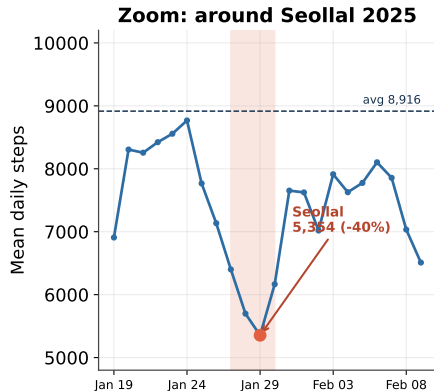
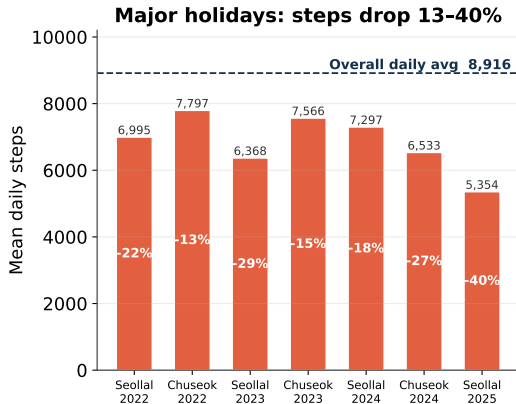
Despite a lower sign-up rate, **men average ~15% more steps** (9,238 vs 8,014/day). Steps rise with age to a peak in the **60s**, easing slightly at 70+.

## Weekly rhythm: a pronounced weekend slump



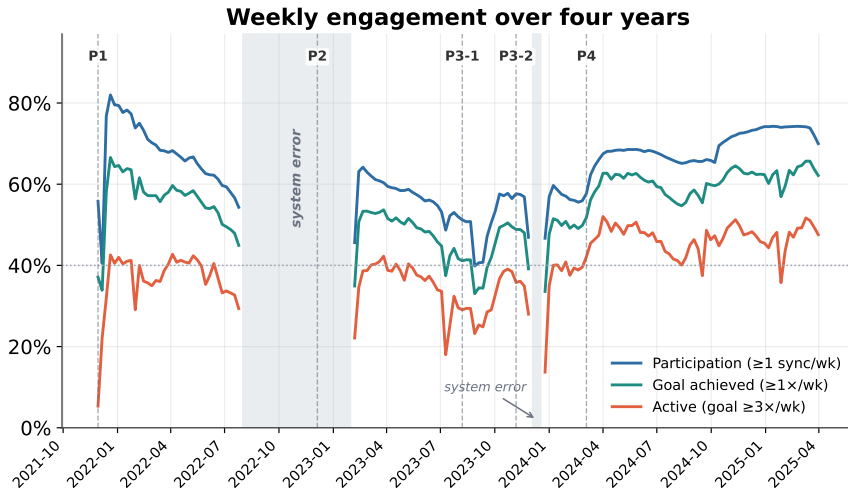
Activity is **weekday-anchored**: Sunday steps fall **15.7%** below the weekday average. Participation and goal achievement dip on weekends too.

## Holidays: the steepest dips of all



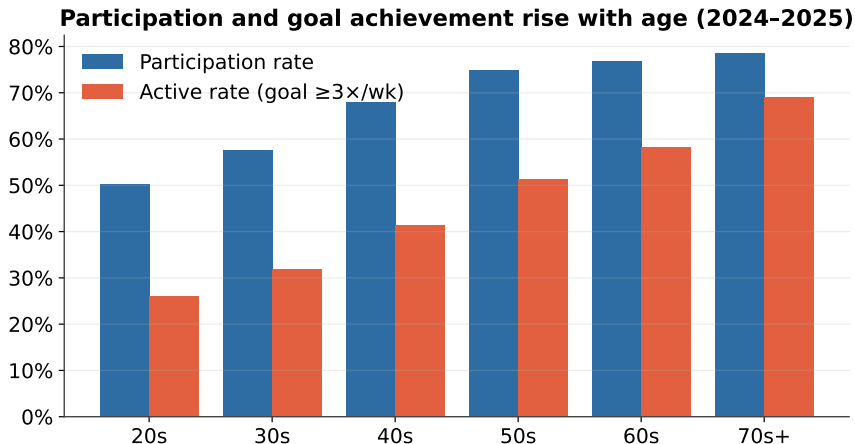
Korea's major holidays — **Seollal** (Lunar New Year) and **Chuseok** — bring far larger drops than weekends: **13-40%** below the overall daily average on the holiday itself, with Seollal 2025 the single lowest day in the record (5,354 steps, -40%).

## Engagement over time: ~40% stay “active”



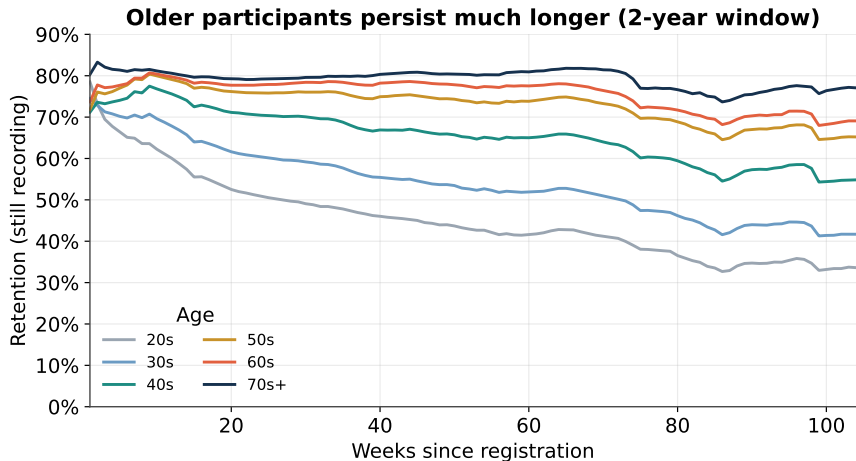
After each enrollment wave, engagement settles into a steady regime: ~70% participate weekly, ~60% hit the goal weekly, and ~40–45% remain active ( $g \geq 3/\text{week}$ ).

## Engagement rises steadily with age



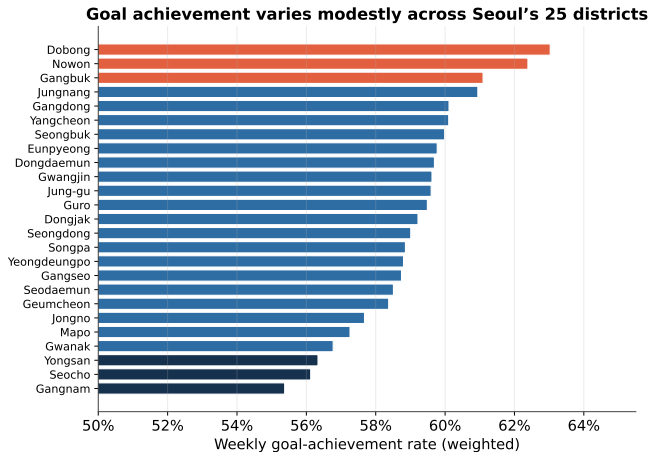
Older members are *more* engaged: both participation and the active rate climb monotonically from the 20s to the 70s. Younger members are the hardest to retain.

## Retention: older participants persist far longer



Two years after joining, retention ranges from **~4%** (20s) to **~24%** (60s). Persistence increases with age across the entire follow-up.

## Regional variation is modest across Seoul



Weekly goal-achievement spans only ~55–63% across all 25 districts — engagement is fairly even city-wide, with a mild edge for several outer-northern districts.

## Findings & implications

---

1. **Scale works.** A municipal mHealth program reached **2.18M** members, with a **1.6M** active base.
2. **Sign-up  $\neq$  intensity.** Women join more (63%), but men average  $\sim 15\%$  more steps.
3. **Age is protective.** Engagement, goal achievement and retention all **rise with age**; younger members churn fastest.
4. **Habits are weekday-anchored.** A consistent  **$-15.7\%$**  Sunday slump.
5. **Activity sustains.**  $\sim 40\text{--}45\%$  stay active ( $g \geq 3/\text{week}$ ) in steady state.

## Policy already moved this way

- Late 2025 (after our data window), the program added a **weekend rule**: miss the step goal on *both* weekend days and the **weekly bonus is withheld**.
- This directly targets the **weekend slump** we document — our descriptive finding aligns with a real policy change.

## Other design levers

- Protect early retention of **younger / short-tenure** members.
- Leverage older adults as a built-in retention success.

## Future analysis

- **Link step data to health records.** We used step data only; joining clinical / check-up data at scale could test whether *sustained* activity improves health — now feasible as the panel lengthens (early waves were too short before).
- **Mental-health outcomes.** Higher daily step counts are linked to fewer depressive symptoms (e.g. +1,000 steps/day  $\approx$  9% lower depression risk; Bizzozero-Peroni et al., *JAMA Netw Open* 2024) — a natural extension for this population.
- **Effectiveness & equity:** pre/post step change; deeper district / enrollment-type analysis.

# Thank you

If you want more detail, please reach me at

**ydjeong@uos.ac.kr**

**Youngdoo Jeong • Dept. of Statistics, University of Seoul**

**JSM 2026 • Health Policy Statistics Section**

This research was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2022-NR068754).